

MEBC 2026 – Proof of Concept (PoC) Video Requirements

This document sets out the complete Proof-of-Concept (PoC) video requirements for all three classes of the Monaco Energy Boat Challenge 2026: Energy Class, AI Class, and SeaLab Class.

The purpose of the PoC video is to demonstrate that each team has achieved a **functional prototype** or a **fully functional system test** before the event.

Teams must upload their PoC video to the Team Dashboard no later than the **22nd of May 2026**, and **notify the organiser by email** immediately after submission.

GENERAL EXPECTATIONS FOR ALL CLASSES

Teams are encouraged to submit their PoC videos well ahead of the deadline. Early submission is preferred and allows the Technical Committee to review this content and provide guidance where needed. If a team foresees that they will not be able to meet the PoC video deadline, they must provide a written, detailed explanation **at least 2 weeks before the deadline**, and make a proposal of the next steps. The Technical Committee will then evaluate this proposal and inform the team by email regarding the decision. **Last-minute submissions will not be accepted if the team is unable to provide the required PoC video evidence before the communicated deadline.**

The PoC video must be **clear, continuous, and strictly technical**. It should show the real system in operation without cinematic edits, music, or scenes that do not contribute to the evaluation of the boat or its systems. Only the necessary technical content should be included. The video can contain live commentary or a written explanation, for which the transcript may be added separately.

ENERGY CLASS REQUIREMENTS

Each Energy Class team must demonstrate a functional propulsion system either on the water or on a fully assembled test bench.

1. On-Water Demonstration

The preferred option is a working boat operating on water under its own propulsion.

The video must show:

- The boat floating and moving.
- All main components installed on the boat (energy source, motor, inverter, steering system, cockpit module).
- A full operational sequence: throttle movement, corresponding motor reaction, propeller rotation, acceleration, deceleration, and steering response.

2. Component Substitution Option

If supply-chain or manufacturing constraints prevent the installation of new components, the team may use main components from previous years. This option is accepted to prove integration capability and system understanding. The team must explain which components are substituted and why.

3. Test Bench Demonstration

If the boat cannot be tested on water, the PoC video may instead show a fully assembled system test bench. The chain from throttle input to rotating propeller must be demonstrated. All major components must be physically present, connected, and operational. Where applicable, the team should also show safety-related functions. The cockpit is still subject to demonstration with most of its mechanical elements.

AI CLASS REQUIREMENTS

AI Class teams must meet all Energy Class requirements plus additional demonstrations related to autonomous navigation. The capabilities of the vessel can be demonstrated by a pilot, or a remote operator.

1. **Sensor Demonstration**

The video must clearly show the sensors mounted on the boat or test bench, including all used sensors like cameras, lidar, radar, GPS, RTK correction system, etc., and the required dual steering-position sensors.

2. **Data Visualisation and Perception**

Teams must present their AI visualisation interface, displaying (real-time, or replayed) perception data such as object detection, labelled bounding boxes, distance estimation, point clouds, and localisation information etc.

3. **Remote Control and Autonomous Mode Switch (AMS)**

The PoC video must include a demonstration of the AMS system, covering the transition between manual and autonomous modes, remote activation, system checks prior to mission submission, and emergency stop capability. If possible, the team must show how the system behaves in communication loss scenarios.

4. **Test Bench Option**

If the AI system cannot be fully mounted on the vessel yet, a complete bench-test demonstration of sensors, actuators, and autonomous processes is acceptable. The crucial point is that the autonomy chain must be functional. The video should show the vessel and explain why it is not functional.

SEALAB CLASS REQUIREMENTS

SeaLab Class teams must meet all Energy Class requirements and must also record and present specific system measurements during two recommended operational test cases. Test bench demonstrations are only acceptable in specific cases if the team can ensure a working prototype by the event. These tests may be conducted on the water or on a fully integrated test bench.

1. Test Case 1 – Mid-Power

Rest – System Energised – Acceleration – 10 minutes at constant mid-power –
Deceleration – 5 minutes rest – System de-energised.

2. Test Case 2 – Maximum Power

Rest – System Energised – Acceleration – 10 minutes at high / maximum constant power –
Deceleration – 5 minutes rest – System de-energised.

Measurements must include:

- Battery signals (recommended): e.g. time, pack and module voltages, current, temperatures, state of charge, isolation resistance, and isolation status.
- BMS signals (if available): contactor states, over-/under-voltage and over-current flags, and temperature warnings.
- Cooling system parameters (if available): temperatures, pressures, and pump flow rates.
- Fuel cell data (if applicable): e.g. voltage, current, temperatures, fuel rate, fuel gauge, pump pressure and flow.
- Inverter parameters (if available): voltages, currents, fault states, and coolant data.
- Motor parameters (recommended): voltage, current, torque, rotational speed, temperature data, and coolant parameters.
- Transmission data (if available): torque, rotational speed, and temperatures.

The video may show these measurements as screen recordings, overlays, or footage of the live data interface, in a 1-2 min video. Log files should be submitted as supporting documentation of the full measurement. Measurements can be made on a testbench environment.

WHAT TO SUBMIT

Each team must upload:

- **The complete PoC video.**
- **A confirmation email to the organiser once the upload is complete.**
- Relevant data logs or screenshots, if applicable.
- A short-written summary if needed, explaining the setup, any substitutions, and whether the system shown is a full or partial demonstration.

FINAL REMARKS

The PoC video is a mandatory part of the registration process. Teams that do not submit an adequate video may not be accepted into the final stage of the competition. The PoC video is reviewed for completeness, clarity, and alignment with the technical regulations but **does not replace the on-site technical inspection in Monaco.**